

Curriculum Vitae

Tewodros Amberbir Habtegebrial

ML ENGINEER · SNAP INC.

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Education

RPTU Kaiserslautern-Landau

Kaiserslautern, Germany

PHD, COMPUTER SCIENCE

Jan. 2018 - Apr. 2025

- Advisor: Prof. Didier Stricker
- PhD Thesis: "Learning View Synthesis from Minimal Scene Specifications".
- Grade: Summa cum laude

RPTU Kaiserslautern-Landau

Kaiserslautern, Germany

MSc, COMPUTER SCIENCE

Nov. 2014 - Jan. 2017

- Advisors: Dr. Christian Bailer, Prof. Didier Stricker
- Thesis: "Novel-view Synthesis with Deep Convolutional Networks".
- Specialization: Intelligent Systems
- Overall Grade: 1.3

Mekelle Institute of Technology

Mekelle, Ethiopia

BSc, ELECTRONICS AND COMMUNICATIONS ENGINEERING

Oct. 2007 - Aug. 2012

- Advisor: Dr. Workineh Gebeyehu
- Thesis: "Visual Object Tracking Using Particle Filtering".
- Overall Grade: 3.79/4.00

Awards & Fellowships

Google PhD Fellowship

2021-2023

GOOGLE RESEARCH

Research area: Computer Vision

CS Graduate Scholarship

2015-2021

RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Graduate Scholarship to pursue Master in Computer Science

Merit Scholarship

2008-2012

MEKELLE INSTITUTE OF TECHNOLOGY

Mekelle, Ethiopia

Professional Experience

Machine Learning Engineer

SNAP INC

Sep. 2024 - present

Working on Machine Learning and Computer Vision

Researcher

Kaiserslautern, Germany

GERMAN RESEARCH CENTER FOR ARTIFICIAL INTELLIGENCE (DFKI)

Jan. 2017 - Nov. 2023

Research projects on Image-based Rendering, VR and 3D Reconstruction

Software Engineer

Addis-Ababa, Ethiopia

ICOG LABS

Oct. 2013 - Oct. 2014

I worked on research projects in unsupervised learning from videos and single image collections

Assistant Lecturer

AKSUM UNIVERSITY

Network Analysis and Synthesis, Electromagnetic Field Theory

Aksum, Ethiopia

Aug. 2012 - Sep. 2013

Internships and Student Researcher Positions

Google Research

STUDENT RESEARCHER: WITH DR. STEFAN POPOV

- Topic: "Multi-view Consistent Diffusion Models"

Remote

Aug. 2022 - Mar. 2023

Meta Reality Labs

INTERN: WITH DR. CHAKRAVARTY CHAITANYA, DR. FEDERICO PERAZZI

- Topic: "Low-Light NeRF"

Remote

Jun. - Sep. 2021

Google Summer of Code

WITH OPENCOG PROJECT

Summer 2015

Teaching Experience

3D Computer Vision

RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Winter Semesters 2018-2022

Graph Theory

RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Summer Semester 2016

Case-based Reasoning

RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Winter Semester 2016

Programming Languages

Python, Lua, Matlab, C, C++

CUDA, PyCUDA

Selected Publications

Text2MPI: Learning to Generate Multi-Plane Images from Text. Tewodros **Habtegebrial**, Stefan Popov, Didier Stricker, Varun Jampani and Vittorio Ferrari. Under Review, 2024.

SOMSI: Spherical Novel View Synthesis with Soft Occlusion Multi-sphere Images. Tewodros **Habtegebrial**, Christiano Gava, Marcel Rogge, Stricker Didier and Varun Jampani. CVPR, 2022.

Generative View Synthesis: From Single-view Semantics to Novel-view Images. Tewodros **Habtegebrial**, Varun Jampani, Orazio Gallo, and Didier Stricker. NeuRIPS, 2020.

Fast View Synthesis with Deep Stereo Vision. Tewodros **Habtegebrial**, Kiran Varanasi, Christian Bailer, and Didier Stricker. VISAPP, 2019.

Deep Convolutional Networks For Snapshot Hyperspectral Demosaicking. Tewodros **Habtegebrial**, Gerd Reis, and Didier Stricker. WHISPERS, 2019.

Fast Feature Extraction with CNNs with Pooling Layers. Christian Bailer, Tewodros **Habtegebrial**, Kiran Varanasi, and Didier Stricker. BMVC, 2018.

SphereGlue: Learning Keypoint Matching on High Resolution Spherical Images. Christiano Gava, Vishal Mukunda, Tewodros **Habtegebrial**, Federico Raue, Sebastian Palacio and Andreas Dengel. CVPR-Workshops, 2023

Amharic Text Image Recognition: Database, Algorithm, and Analysis. Birhanu Hailu Belay, Tewodros **Habtegebrial**, Marcus Liwicki, Gebyehu Belay and Didier Stricker. International Conference on Document Analysis and Recognition (ICDAR), 2019

Factored Convolutional Neural Network For Amharic Ccharacter Image Recognition. Birhanu Hailu Belay, Tewodros **Habtegebrial**, Marcus Liwicki, Gebeyehu Belay and Didier Stricker. International Conference on Image Processing (ICIP), 2019